The Magnetization of Red Hill Dike, Tasmania

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Abstract. A detailed laboratory study has been made of the magnetization of Red Hill dike, which is a single rock body within the Tasmanian dolerite intrusive complex. A potassium-argon age of 170 m.y. has been measured on this dike using samples from some of the same sites at which the magnetization was measured. The results show that the direction of natural remanent magnetization has an insignificant amount of secondary magnetization and that there is no evidence of its having any systematic error due to stress or shape effects. The mean direction is $D_m = 294°$, $I_m = -75°$, which is regarded as a good estimate of the geomagnetic field 170 m.y. ago; the steep inclination is similar to that found in other Jurassic rocks from Australia.

Introduction. The results of a regional survey of the magnetization of the Tasmanian dolerites have been given in the preceding paper; in this paper the evidence from a detailed laboratory study of a single rock body is described.

Samples were collected from a large, dikelike intrusion at Red Hill, some 15 to 30 km SSW of Hobart, in southern Tasmania. This has been intensively studied petrologically by *McDougall* [1961, 1962], and it is from here that rock samples were collected by him for his K-Ar age determinations. The dolerite of the Red Hill area is intrusive into flat-lying Permian and Triassic sediments (mainly into the Permian beds). The form of the intrusion is essentially that of a dolerite sheet about 400 meters thick, which is gently transgressive upward through the Permian sequence from east to west. The Red Hill dike is up to 1500 meters wide and extends almost vertically upward from the underlying sheet for over 350 meters. It can be traced north and south of the area studied for a total length of over 10 km. The field relations suggest that the intrusion was initially roofed by sediments. A geological sketch and schematic cross sections are shown in Figures 1 and 2.

The dolerite magma was liquid at the time of emplacement, except for a few per cent of microcrystals of pyroxene, and at the contacts with the intruded sediments it was chilled to a very fine-grained rock, or in some cases to a glass. The dolerite is tholeiitic, and is similar to the Mesozoic dolerites of the Karroo, Brazil, Antarctica, and Palisades. During cooling and crystallization, marked differentiation took place. Progressively more silicic quartz dolerites occur along a traverse from the chilled contacts in the lowest exposed section toward the central parts of the Red Hill dike; the dolerite grades into fayalite granophyre and finally into silicic granophyre between the center of the dike and the summits of Red Hill and O'Briens Hill. These differentiates are markedly enriched in iron, silica, and alkalies and are impoverished in magnesium, calcium, and aluminum, in comparison with the original magma. The differentiation in the Red Hill dike is more extreme than that previously found by *Edwards* [1942] in his study of several other Tasmanian dolerite intrusions.

From the geological evidence, a Jurassic age had been considered the most probable [*Banks*, 1958], and this agrees with the age of 170 m.y. determined more recently by *McDougall* [1961] from K-Ar measurements on plagioclase and pyroxene.

The magnetic minerals series form from about 1 to about 10 per cent of the rock, by volume. The magnetic mineral seems uniform throughout the dike [*McDougall*, 1962, p. 293] and consists of about 50 per cent of nearly pure magnetite and 50 per cent of slightly magnetic

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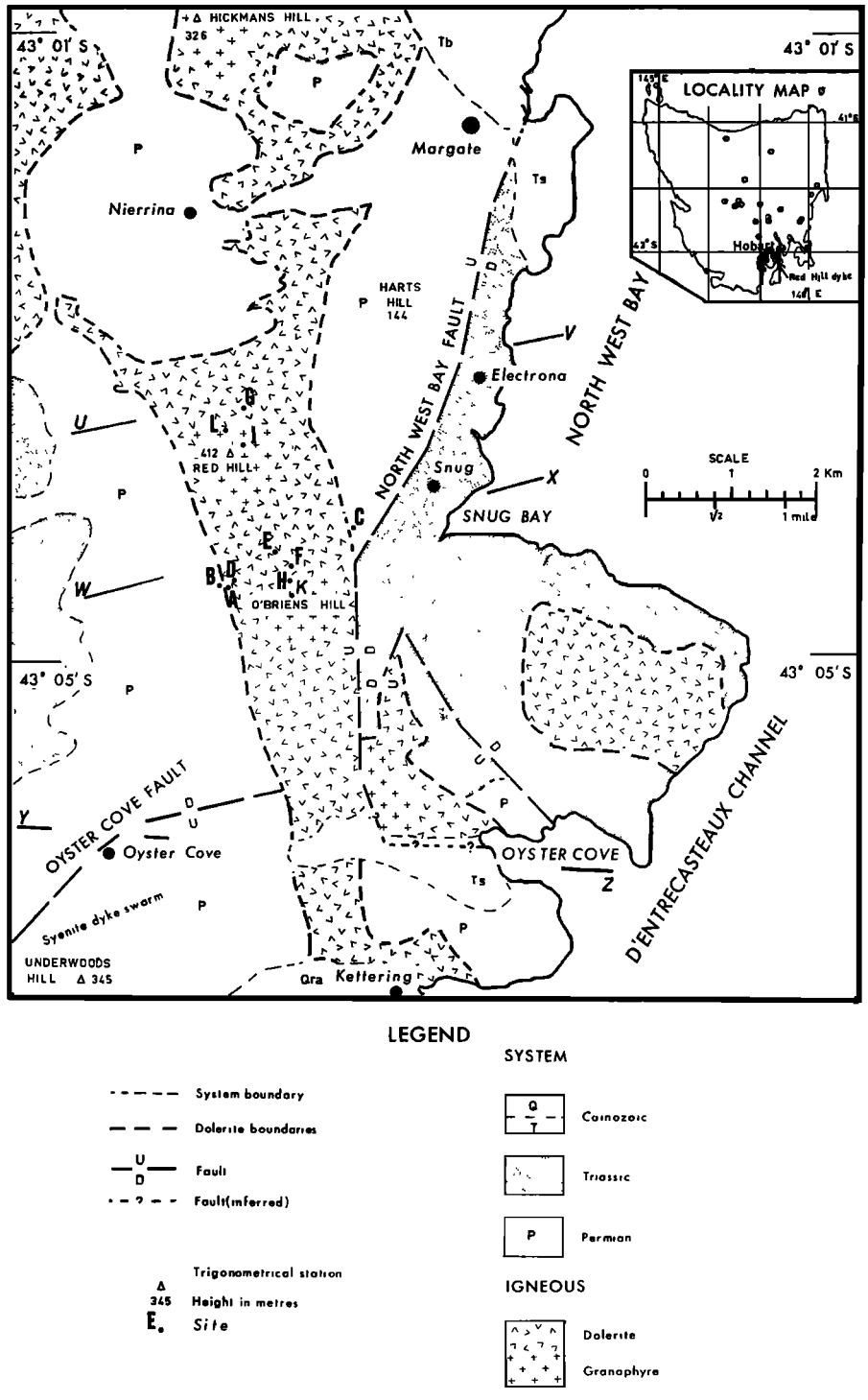


Fig. 1. Geological sketch showing collecting sites, Red Hill area, Tasmania. Inset locality map indicates collecting sites of *Irving* [1963]. Lines *UV* and *WX* show lines of sections, illustrated in Figure 2.

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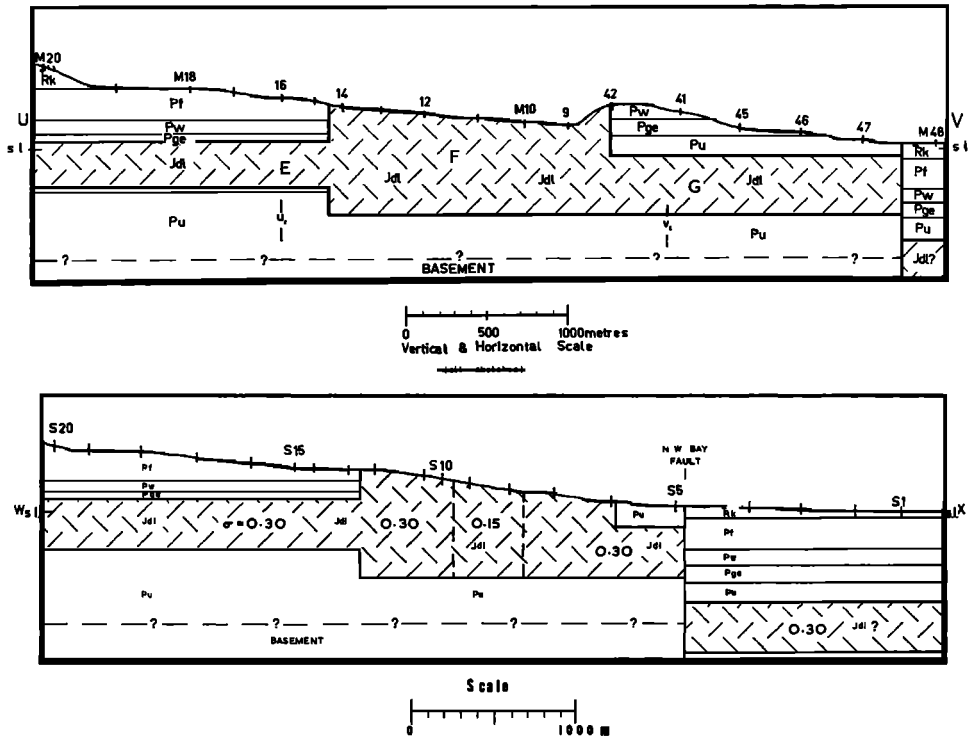


Fig. 2. Schematic cross sections across the Red Hill dike. Lines *UV* and *WX* are shown in Figure 1. Sections are based on geological mapping and reconnaissance gravity traverses near the section lines (see *McDougall and Stott* [1961]; this figure is taken from Figures 2 to 4 of their paper).

ilmenite, occurring as separate exsolution lamellas.

Results. Samples of the various rocks have been collected from nine sites within the dike and from baked sediments from the contacts at both sides of the dike. The sites are shown in Figure 1 and site descriptions are given in Table 1.

Mean directions of natural remanent magnetization (NRM) at each site are given in Table 2 and plotted in Figure 3. The direction at site L diverges from the main group, and the over-all mean direction has been calculated both with and without the result for this site. The baked sediments at site B are weakly magnetized, but they give a good cluster of directions that are in agreement with the dolerite (Figure 4). At site C the remanent magnetization is extremely weak, and this, combined with a high susceptibility, makes accurate measurements impossible.

The directions resulting from demagnetizing the NRM of some specimens at five sites are summarized in Table 3. The variations in intensity of magnetization when the NRM is subjected to increasing alternating fields are shown in Figure 5 for the five sites D, E, F, J, and K. These curves show, for successive demagnetization, the ratio of the remaining intensity to the initial intensity M_n and the peak alternating field (H_p~) used. At sites D, F, and J, similar measurements were made when thermoremanent (TRM) and isothermal (IRM) magnetizations were being demagnetized, and the results are shown in Figure 6. The curves for NRM and TRM are similar. Table 4 shows the actual intensities of magnetization (NRM and TRM) during these experiments. At site D (20 meters from the west contact) the TRM in a 0.6-oe field is about 15 times greater than the NRM, at site F (center of the dike) $M_{t(0.6)}$ is only about twice the value of M_n, and at site J the values are almost equal. The magnetization at site J is very large, and comparison of demagnetization curves (Figure 6) shows that the

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TABLE 1. Site Descriptions

Position	Rock Description
Site A:	
West contact, bed of Snug River, left bank. 0 to 1 m from contact. (Altitude: 70 m)	Chilled dolerite, usually very fine-grained holocrystalline, sometimes devitrified glass. Grain size 0.01 mm; small amount iron ore as crystals (see M172).*
Site B:	
Same site as A, 0 to 2 m from contact (70 m)	Baked sediments.
Site C:	
East contact, from roadside, about 15 m above bed of Snug River (70 m).	Baked sediments.
Site D:	
About 15 to 30 m from West contact, in bed of Snug River (70 m).	Fine- to medium-grained quartz dolerite. Iron ore about 2 per cent (M212).
Site E:	
Melville Ck, (N. side of Snug River valley) (about 65 m).	Medium-grained quartz dolerite (between M9 and M209).
Site F(i):†	
Center of Red Hill dike in bed of Snug River (about 65 m).	Coarse-grained quartz dolerite (M210).
Site F(ii):†	
Center of dike, on roadside on N. flank of O'Briens Hill (140 m).	Coarse-grained quartz dolerite (M221).
Site G:	
Poverty Gully Ck., north of Red Hill (175 m).	Fayalite granophyre (M384).
Site H:	
Center of dike on N. flank O'Briens Hill (230 m).	Fayalite granophyre, just above transition from quartz dolerite (M223).
Site J:	
Summit of Red Hill (410 m).	Granophyre. Up to 10 per cent iron ore (M19).
Site K:	
Near summit of O'Briens Hill (310 m).	Granophyre. About 5 per cent iron ore (M176).
Site L:	
N. flank of Red Hill. About 400 m N.W. of summit (275 m).	Granophyre. About 2 per cent iron ore (M162).

* Numbers 'M172,' etc., refer to specimen numbers in *McDougall* [1960, 1962], where detailed descriptions can be found. The specimens referred to are from the same collecting locality.

† Sites F(i) and F(ii) are taken together as site F.

NRM is indistinguishable from the TRM (0.6); the high intensities are due to the very high iron ore content (up to 10 per cent of ore minerals).

Results of experiments in which rock specimens acquired TRM and IRM in the laboratory while subject to directed stresses of 500 to 1000 kg/cm^2 have been reported earlier [*Stott and Stacey*, 1960]. Those results relating to dolerite samples are groups 3, 5, and 6 in Table 1 of that paper (group 5 is from site E of this paper). The directions in stressed and unstressed specimens were parallel to the applied field.

Measurements of susceptibility (in a field of about 0.5 oe) in three orthogonal directions showed no significant differences. This apparent magnetic isotropy is confirmed by measurements made in a field of about 10,000 oe by *Stacey* [1960*a*]. This is a more sensitive method, and Stacey found an anisotropy of less than 5 per cent; this slight anisotropy is directed randomly from sample to sample. (The anisotropy here is the proportionate difference in the axes of the equivalent magnetic ellipsoid.) *Stacey* [1960*a*] shows that this anisotropy should have no measurable effect on the direction of remanent magnetization, and his finding is confirmed by measurements made on specimens acquiring TRM in the laboratory as follows: the mean inclination of 12 specimens cooled from 600°C to room temperature in the earth's magnetic field is −65°, compared with the local field inclination of −65.6°.

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TABLE 2. Mean Site Directions of Magnetization (NRM)

Site	S	N	D_m	I_m	R	k	a_{95}
				Dolerites			
A	3	10	301	−75	9.75	36	8
D	7	25	321	−84	24.83	140	2
E	5	9	279	−77	8.96	202	4
F	7	13	287	−70	12.92	16	3
G	2	5	283	−66	4.98	159	6
H	6	12	302	−79	11.89	100	4
J	3	9	299	−86	8.78	36	9
K	4	9	299	−58	8.86	57	7
L*	3	8	38	−58	7.88	57	7
				Baked Sediments			
B	6	23	310	−78	22.60	55	4
C	4	12	random		2.4		
Mean dolerite (9 sites)			308	−77	8.70	7	10
Dolerite excl. L (8 sites)			294	−75	7.90	70	7

At each site, N specimens cut from S samples are measured; the mean directions (D_m, I_m) are calculated by Fisher's method, giving each specimen unit weight. R is the magnitude of the resultant vector; k the estimate of precision; a_{95} the semiangle of the cone about the mean, in which the true mean direction lies with $P = 0.95$; over-all mean calculated giving each site unit weight.

*Site L is excluded on bottom line; see text.

Discussion. The mean directions of magnetization at the nine dolerite sites are in good agreement with each other, except for site L. The direction of site L makes an angle of 34° with the mean of the nine sites, and, if these nine results represent the same formation direction, the probability of this occurring by chance is only $P = 0.001$. It seems likely that the outcrop at this site has moved in a large landslip at some time since it was originally formed. The mean direction of the remaining eight sites makes an angle of 21° with the present earth's field and of 30° with the dipole field. The probability that the true mean direction is that of the present field is $P \approx 10^{-18}$; the probability that it is that of the dipole field is even smaller.

Sites A and D to K represent an ordered sequence of rock types from the fine-grained, almost glassy rock of site A to the very coarse granophyre at sites J and K. The good agreement of directions is shown in Table 2 and Figure 3. For example, the angle between directions in the extremes of rock types is $\angle$ (A − J) = 11°, $\angle$ (A − K) = 17°. Even if we assume that there is some common chemical or mineral factor that causes all these results to have a systematic error, it is unlikely to apply to the baked sediments (site B), where the mean direction agrees with that measured in the dolerite. In particular, let us compare the results at the adjoining sites A and B (all samples are from within 2 meters of the contact, site A in dolerite and site B in the baked sediments); the mean directions agree to within 5°, which is less than the statistical error in the determinations. One specimen core crosses the actual contact, and the two halves (the core broke along the contact when being removed) have directions of magnetization within 2° of each other.

The demagnetization results show, first, that the NRM of the rocks is due almost entirely to a single stable component of magnetization and, second, that in all specimens studied the demagnetization curves of NRM and TRM are very similar. In one case, site J, the magnitude of the NRM is equal to that of the TRM that was acquired in the laboratory in a field of 0.6 oe. Five sites have been studied in detail, and the mean direction of NRM at these sites is (296, −76); after magnetic cleaning in 150 oe, when

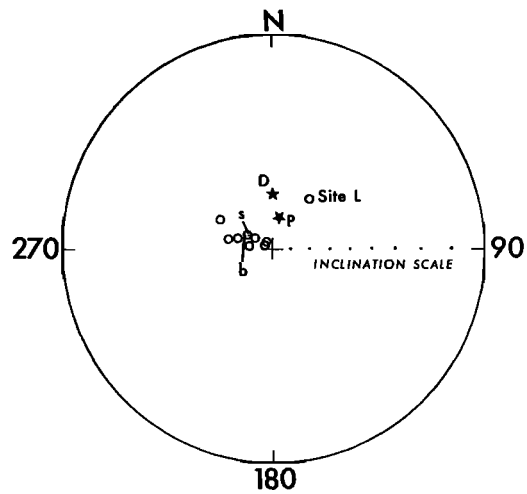


Fig. 3. Site mean directions at the nine dolerite sites. The mean direction using sites (s), using specimen directions (b), the dipole field direction (D), and present field direction (P) are also shown; equal-angle (Wulff) projection.

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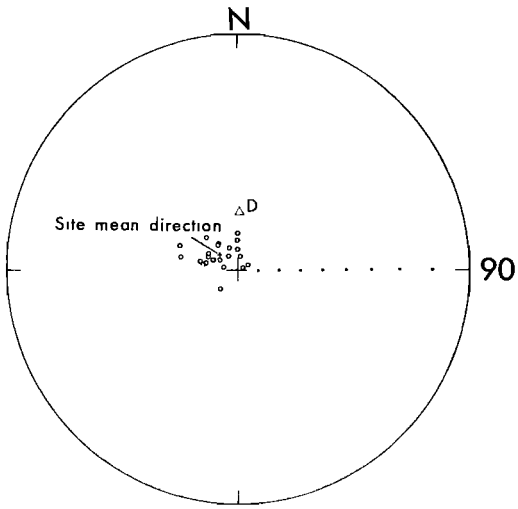


Fig. 4. Specimen directions at site B, sediments baked by the dolerite intrusion. Dipole field direction (D) and site mean direction are shown.

the dispersion is least, the mean direction is (282, −74), a change of only 3°. Even after severe demagnetization in a peak field of 750 oe, the mean direction is not changed significantly, although the results are more scattered.

The similarity between the demagnetization curves for NRM and TRM, and the agreement in direction between sites A and B and between B and the dolerite as a whole, support the view that the NRM is almost entirely due to TRM acquired in the prevailing field at the time of intrusion. The intensity of TRM in the present-day field of samples from site J, compared with their NRM, shows that the field at that time was not weaker than now.

The question, then, is whether or not the magnetization was parallel to the field in which the rock cooled. The suggestion that magnetostriction might cause a systematic deflection of the magnetization [*Graham*, 1956] is not very likely. The experimental evidence shows that no detectable deflection occurs in Tasmanian dolerite, as was concluded by *Stott and Stacey* [1960].

It is worth considering the general question of magnetostriction very briefly, because several papers have appeared recently and the position has changed within the last two years or so. Following the original suggestion of *Graham* [1956], two types of experiments have been used in investigating the effect. *Graham et al.* [1957] used specimens of Adirondack gneiss, which were subjected to uniaxial stress and the direction of magnetization measured with the load applied, at room temperature; they found quite large changes in the magnetization. This deflection, due to isothermally applied, directed stress is essentially a piezomagnetic effect, and *Stacey* [1958] pointed out that the results, although they could be explained theoretically, were not directly applicable to TRM *acquired* under non-hydrostatic pressure.

A second type of experiment was devised by *Stott and Stacey* [1960], as mentioned earlier, in which specimens acquired TRM and IRM under directed stress of up to 1000 kg/cm² and the magnetization was measured *after* the load was removed. This experiment more nearly repro-

TABLE 3. Site Mean Directions during Magnetic Cleaning at Five Sites

Symbols are the same as for Table 2, ($N = S$) and w and b are the within- and between-site precisions.

Site	N	D_m	I_m	R	a_{95}	D_m	I_m	R	a_{95}	D_m	I_m	R	a_{95}
	Field:		NRM				75 oersteds				150 oersteds		
D	5	318	−84	4.96	8	289	−81	4.98	5	286	−79	4.98	5
E	5	286	−78	4.99	4	281	−76	4.95	8	298	−76	4.95	8
F	6	286	−70	5.97	5	287	−72	5.97	5	288	−71	5.94	8
J	3	308	−85	2.92	25	272	−79	2.89	30	270	−75	2.90	28
K	4	301	−61	3.96	11	288	−66	3.94	13	273	−67	3 96	11
	5 Sites:	296	−76	4.93	10	284	−75	4.98	6	282	−74	4.98	5
Mean													
	23 Specs:	295	−75	22.5	4	285	−75	22.6	4	284	−74	22.6	4
k			60				173				204		
w			88				68				67		
b			79				470				705		

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TABLE 3. (Continued)

Site	N	D_m	I_m	R	a_{95}	D_m	I_m	R	a_{95}	D_m	I_m	R	a_{95}
	Field:		300 oersteds				450 oersteds				750 oersteds		
D	5	257	−81	4.96	7	221	−76	4.80	18	228	−67	4.56	27
E	5	291	−68	4.29	35	319	+48	2.43	..	118	+7	0.8	..
F	6	286	−68	5.78	14	279	−75	5 72	16	300	−40	3.21	73
J	3	299	−84	2.88	31	131	−78	2.94	22	71	−63	2.35	87
K	4	265	−70	3.86	20	271	−53	3.05	62	227	−73	2.95	67
Mean	5 Sites:	280	−75	4.96	8	292	−67	3.59	56	161	−83	3.32	64
	23 Specs:	280	−74	21.6	8	281	−72	15.4	22	256	−78	11.3	32
k			89				2.8				2.4		
w			15				4.4				2.0		
b			∞				6.7				32		

duces natural events, and specimens of various igneous rocks from Australia were examined. The result, that no significant deflection occurs, left the general problem unresolved, because at that stage only these two experiments, in which different rocks were used, had been reported. Recently, however, *Kern* [1961] measured TRM acquired under load, after the load was removed (the second type of experiment), in the *same* rocks used by Graham et al. for their experiment, and found no deflection due to uniaxial stress, except in one very anisotropic rock. This adds weight to the idea put forward by *Stacey* [1960*b*] that magnetization acquired under load is deflected from the direction of the applied field and *then returns* to this direction when the directed stress is removed. This explains all the observed facts about the magnetization of *isotropic* rocks in relation to uniaxial stress.

In view of Kern's work, there is no experimental evidence that rocks of any type normally used for paleomagnetic work suffer appreciable, permanent deflection of their magnetization by stress, although there is some evidence of small deflections in anisotropic rocks [*Hall and Neale,* 1960; *Hall,* 1961; *Stott and Stacey,* 1961].

Strangway [1961] suggests that the magneti-

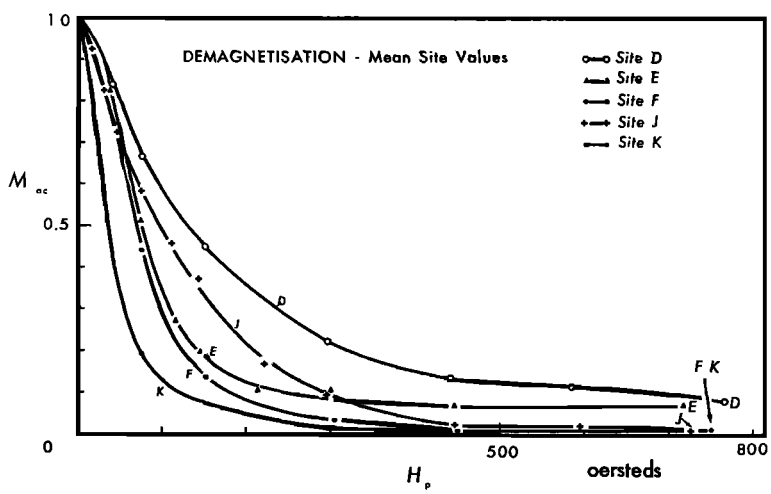
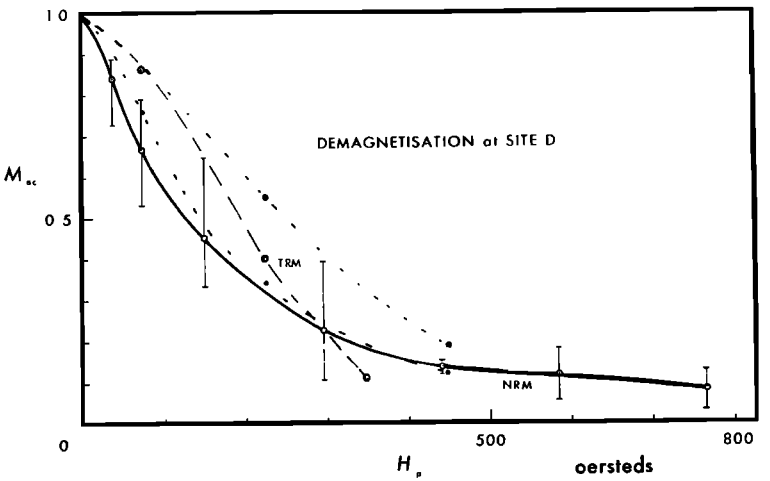
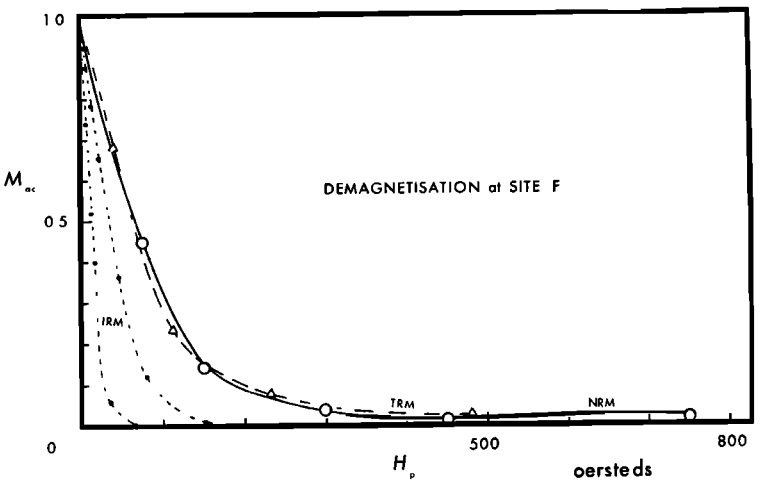


Fig. 5. Mean values of the relative intensity of NRM after demagnetizing in increasing alternating fields at sites D, E, F, J, and K. The value M_{ac} on the vertical axis is the ratio of remaining to initial intensity (M/M_n) after treatment in the corresponding peak alternating field ($H_p\sim$) shown along the horizontal axis.

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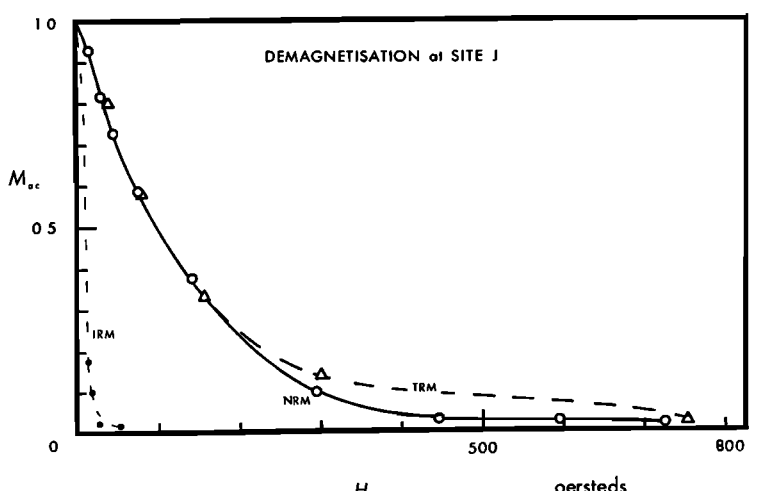


Fig. 6. Comparison of the normalized intensities of TRM in 0.6 oersted, IRM, and NRM at sites D, E, and J during demagnetization. The curve for NRM is solid. The TRM and IRM curves are dashed and labeled. The unlabeled dotted curves at site D are for PTRM acquired between 600° and 500°C (lower curve) and between 400°C and room temperature (upper curve). The NRM values are the mean of three or four specimen disks; the range of values is shown for site D, which is typical. The TRM values are the mean of two specimens at site D and of three specimens at sites E and J. Remaining results are from a single specimen in each case.

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TABLE 4. Intensity of Magnetization of NRM and TRM during Magnetic Cleaning

The figures in parentheses, below the site letter, are the number of specimens (from different rock samples) used in the TRM experiments. The NRM values given refer to the same samples as the TRM values. The first row of figures is the peak alternating field ($H_p \sim$) used in each case.

Site	0	40	75	110	150	225	300	450	750
D NRM	0.10×10^{-3} emu/cm²		0.07		0.05		0.03	0.02	
(2) TRM	1.56		1.39			0.65		0.16	
F NRM	2.93		1.32		0.35		0.07	0.04	
(3) TRM	5.67	3.86		1.32		0.43		0 14	
J NRM	23.7		13.8		8.90		2.30		0.37
(3) TRM	19.7	15.9	11.50		6.64		2.68		0.28

zation of dikes may be deflected toward the plane of the dike. In his examples, the magnetic intensity was high, and there were two magnetic minerals with different Curie temperatures, the component having the lower Curie point being more stable. The latter condition is not present in the Tasmanian dolerite. Most of the NRM withstands heating to 450°C, and there is field evidence that no deflection has occurred. Samples of intrusions used by *Irving* [1963] and for this paper include sills and dikes, and probably they also include bodies of irregular shape; yet the precision of the results is as high as any yet observed in paleomagnetic surveys.

Conclusions. The NRM of the dolerite at the Red Hill dike is due to TRM acquired when the rock cooled (170 m.y. ago). Although magnetic cleaning improves the precision of the result, it does not significantly alter the mean direction of magnetization. There is evidence to suggest that there is no systematic error due to stress or shape, and therefore the direction of NRM is a reasonable estimate of that of the geomagnetic field at the time of intrusion. The time of intrusion (170 m.y. ago) has been measured on the actual rock body sampled for magnetic measurements.

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